

1. The Computing & Processing Core

- **Raspberry Pi Zero 2 W**
 - **Reason:** The central Single-Board Computer (SBC). It hosts the headless Linux environment, handles multi-threaded script execution, processes your offline timers, and executes your local deep learning computer vision models on the edge.
- **MicroSD Card (32GB or 64GB, Class 10)**
 - **Reason:** Acts as the robot's primary hard drive. It stores the operating system file system, Python source code, software dependencies, and your quantized machine learning models.
- **Micro-USB OTG Cable & Mini-HDMI Adapter**
 - **Reason:** Mandatory for the initial standalone setup phase to connect a temporary hardware keyboard and monitor before running the system completely headless.

2. Human-Robot Interaction (HRI) Subsystem

- **GC9A01 1.28-inch Round LCD Display**
 - **Reason:** Functions as the dynamic, digital "face" of the robot. It renders real-time eye expressions via the SPI bus to communicate internal system states to the operator.
- **Raspberry Pi Zero CSI Camera Module (5MP)**
 - **Reason:** The robot's primary vision sensor. It pipes raw, live video data straight to the Pi for real-time localized object tracking and computer vision tasks.
- **Pi Zero Specific CSI Ribbon Cable**
 - **Reason:** Physically bridges the camera module to the uniquely narrow, high-density CSI port on the Raspberry Pi Zero 2 W.
- **MAX98357A I2S DAC Class D Mono Amplifier Module**
 - **Reason:** Decodes digital audio streams straight from the Pi's I2S bus and amplifies them into a crisp analog output, bypassing the Pi Zero's lack of a headphone jack.

- **Stereo Enclosed Speaker (3W, 8Ω)**
 - **Reason:** Serves as the robot's vocal output mechanism. Being fully enclosed ensures maximum clarity and volume for localized text-to-speech feedback and audible status alerts.

3. Mobility & Mechanical Chassis

- **4WD Robotics Rover DIY Wooden Chassis Kit (Base Plate Only)**
 - **Reason:** Utilized purely as the physical mounting deck and structural floor to securely arrange your electronics and hardware modules.
- **DC 12V N20 Micro Metal Gear Motors (200 RPM) [Quantity: 4]**
 - **Reason:** The upgraded drive motors. They deliver incredible physical torque and highly controlled movement via robust, durable internal metal gearboxes within a space-saving footprint.
- **N20 Motor Mounting Brackets (with M2 Screws/Nuts) [Quantity: 4]**
 - **Reason:** Explicitly required to securely clamp the miniature metal N20 motor housings directly onto the wooden chassis plate, preventing them from twisting out of alignment.
- **N20 3mm D-Shaft Rubber Wheels (43mm Diameter) [Quantity: 4]**
 - **Reason:** Specialized tires engineered with a precise 3mm D-shaped inner hub to press-fit tightly onto the N20 metal output shafts without slipping under heavy loads.
- **L298N Dual H-Bridge Motor Driver Module**
 - **Reason:** Bridges the control gap between logic and actuation. It intercepts low-voltage directional commands from the Pi and switches high-voltage battery current directly to your N20 motors.

4. Power & Electrical Distribution (LiPo Overhaul)

- **LiPo Battery 2200mAh 11.1V 3S (Continuous Discharge Rate: 25C)**
 - **Reason:** The primary heavy-duty energy reserve. It supplies raw high-voltage current to the L298N motor driver, allowing your 12V N20 motors to achieve their maximum torque and full 200 RPM speed capabilities.
- **B3 Pro LiPo Balance Charger (2S/3S Compatible)**

- Reason: Safely charges and recalibrates the battery pack by checking and isolating individual cells via the white multi-pin JST-XH balance lead.
- **LiPo Battery Voltage Checker / Low-Voltage Alarm (3S Compatible)**
 - Reason: A vital hardware safety monitor. It plugs directly into the battery's balancing lead to trigger a loud audible alarm the moment a cell drops to unsafe levels, preventing permanent battery swelling or degradation.
- **HW-411A LM2596 DC-DC Step-Down Adjustable Buck Converter (3A Max)**
 - Reason: Essential for system protection. It steps down the raw 11.1V–12.6V from your LiPo battery to a perfectly regulated, safe 5.0V logic rail to run your Raspberry Pi, camera, and screen.
- **Breadboard Half-Size Bare (400 Tie-Points) [Quantity: 2]**
 - Reason: Functions as the main electrical routing matrix to safely split power distribution lines and seamlessly tie all component grounds together into a single common ground.

5. Security Payload & Failsafe Automation

- **5V 1-Channel Optocoupler Relay Module**
 - Reason: Acts as a hardwired, physically isolated electronic power switch. It allows your Python code to completely turn off or power up the transceiver payload via a single digital GPIO command.
- **nRF24L01+PA+LNA Wireless Transceiver Module**
 - Reason: The dedicated radio-frequency hardware payload used for localized network auditing, packet analysis, and signal transmission testing.
- **External SMA Antenna**
 - Reason: Screws directly onto the nRF24L01 module to maximize communication range and maintain clean directional signal isolation during active operations.

6. Structural Hardware & Interconnects

- **Male-to-Female Jumper Wires & Female-to-Female Jumper Wires (20cm)**

- Reason: The nervous system of your robot. These cables route high-speed digital logic communication and power lines smoothly between all separate modules.
- **Nylon Standoffs, Screws, and Nuts Kit**
 - Reason: Crucial for elevating bare printed circuit boards off the chassis floor, cleanly preventing catastrophic short circuits caused by exposed copper tracks touching each other.
- **Double-Sided Foam Tape or Zip Ties**
 - Reason: Securely locks down heavy components—like the 3S LiPo battery pack and the enclosed speaker box—so they do not shift weight during sharp navigational turns.

7. Secondary / Expansion Sensors (For Autonomous Navigation)

- **MPU-9250 9-Axis Attitude Gyro Accelerator Sensor Module**
 - Reason: Delivers real-time accelerometer, gyroscope, and magnetic compass telemetry over the shared I2C bus, giving the local navigation script an absolute heading reference to eliminate calculation drift over time.
- **Ultrasonic Distance Sensor (3.3V Safe HC-SR04 / RCWL-9610)**
 - Reason: Serves as a physical boundary radar, providing immediate distance feedback to the script for automated, non-visual obstacle avoidance.
- **1 kΩ and 2 kΩ 1/4w Film Resistors**
 - Reason: Configured as a precise hardware voltage divider to safely step down the ultrasonic sensor's 5V Echo output to a 3.3V logic level that the Raspberry Pi's pins can safely read without burning out.